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AERO60492 – Autonomous Mobile Robots

Lecture 19 – Rigid body dynamics 2

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Previously

- Understand rigid body dynamics modelling
- Newton-Euler modelling
- Derive generalised equations of motion
- Sources of forces and torques (locomotion)

New aims

- How to use rigid-body-dynamic equations
- Simplifications and linearisations

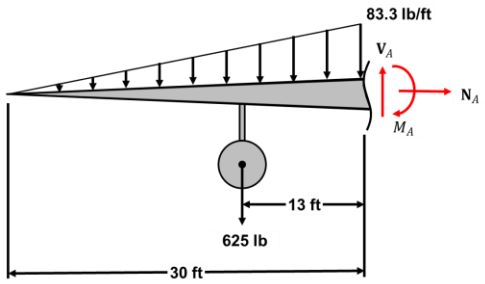
- Can link previous knowledge to form complete motion model of robotic system
- This is the most generalised form – simplifications and caveats exist for many vehicles

Locomotion	+	Rigid body dynamics (lecture 9)	+	Transformations (lecture 7)
How forces and torques are created		Effect on rigid constant mass		Transformation into useful axes for guidance navigation and control

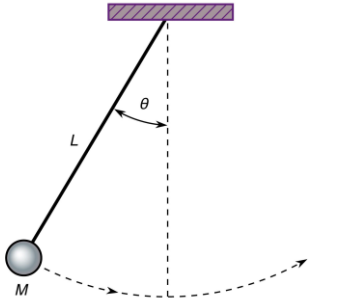
Recap - what is rigid body dynamics

Movement of systems of interconnected bodies under the action of external forces.

Rigid – the bodies are assumed connected by infinitely stiff structure 100% transmission of forces and torques between sub-systems



Dynamics – the science of forces and motion motion



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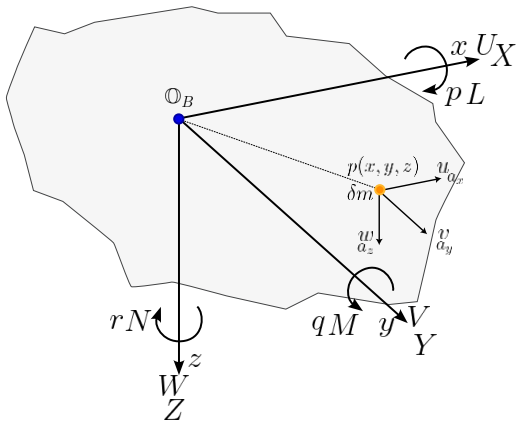
Generalised inertial acceleration

Define a point p within the body reference frame
Point has translational velocity (up, yp, wp) relative to the body
Point has translational acceleration (ax, ay, az) relative to the body
Body has overall velocity and rotational rate (U,V,W) and (p,q,r)

Assume a point within body frame

View looking along z axis

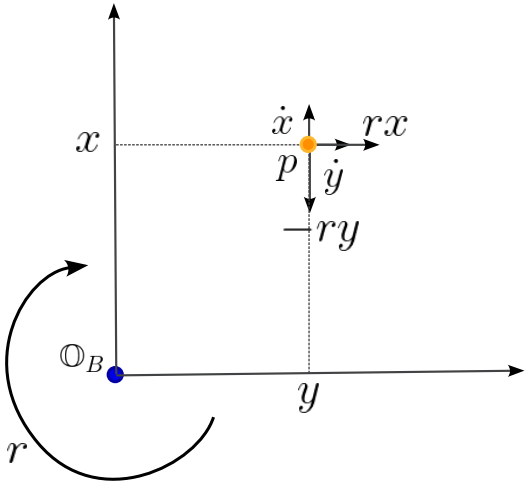
Sum the velocities at point p



$$\bar{p}_p = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\bar{v}_p = \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

$$\bar{a}_p = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix}$$



$$u = \dot{x} + qz - ry$$
$$v = \dot{y} + rx - pz$$
$$w = \dot{z} + py - qx$$

Assumption only made after general derivation $\dot{x} = \dot{y} = \dot{z} = 0$

Generalised equations of motion

$$a'_x = \dot{U} - rV + qW - x(q^2 + r^2) + y(pq - \dot{r}) + z(pr + \dot{q})$$

Motion equation for a massless point (only x shown)

$$\sum \delta m a'_x = X$$

Assume each point p has an incremental mass δm

$$X = \sum \delta m \dot{U} - \sum \delta m r V + \sum \delta m q W - \sum \delta m x (q^2 + r^2) + \sum \delta m y (pq - \dot{r}) + \sum \delta m z (pr + \dot{q})$$

Assume at centre of mass $\sum \delta m x = \sum \delta m y = \sum \delta m z = 0$

Generalised moment equations

$$a'_y = \dot{V} - pW + rU - x(pq + \dot{r}) + y(p^2 + r^2) + z(qr + \dot{p})$$
$$a'_z = \dot{W} - qU + pV + x(pr - \dot{q}) + y(qr + \dot{p}) - z(p^2 + q^2)$$

Motion equations for a massless point (only y and z shown)

$$\sum \delta m(ya'_z - za'_y) = L$$

Assume each point p has an incremental mass δm

$$L = \dot{p}\sum \delta m(y^2+z^2) + qr\sum \delta m(y^2-z^2) + (r^2-q^2)\sum \delta myz - (pq + \dot{r})\sum \delta mxz + (pr - \dot{q})\sum \delta mxy$$

Assume sum of mass at distance squared from centre of mass is Inertia

What is rigid body dynamics

	Translational	Rotational
Principal	$F = ma$	$T = I\dot{\omega}$
Generalised equations of motion	$X = m(\dot{U} - rV + qW)$ $Y = m(\dot{V} - pW + rU)$ $Z = m(\dot{W} - qU + pV)$	$L = I_x\dot{p} - (I_y - I_z)qr + I_{xy}(pr - \dot{q}) - I_{xz}(pq + \dot{r}) + I_{yz}(r^2 - q^2)$ $M = I_y\dot{q} + (I_x - I_z)pr - I_{xy}(qr - \dot{p}) + I_{xz}(p^2 - r^2) + I_{yz}(pq - \dot{r})$ $N = I_z\dot{r} - (I_y - I_y)pq + I_{xy}(q^2 - p^2) + I_{xz}(qr + -dotp) + I_{yz}(pr - \dot{q})$
Vector notation form	$\bar{F} = m\left[\dot{\bar{V}} + (\Omega \times \bar{V})\right]$	$\bar{M} = I\dot{\Omega} + (\Omega \times (I\Omega))$

$$I = \begin{bmatrix} I_x & I_{xy} & I_{xz} \\ I_{xy} & I_y & I_{yz} \\ I_{xz} & I_{yz} & I_z \end{bmatrix}$$

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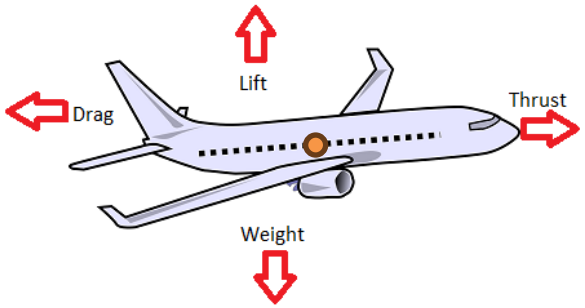
WHY?

Continuous-time equations to predict the motion of a mass subject to forced and moments acting around its centre of mass

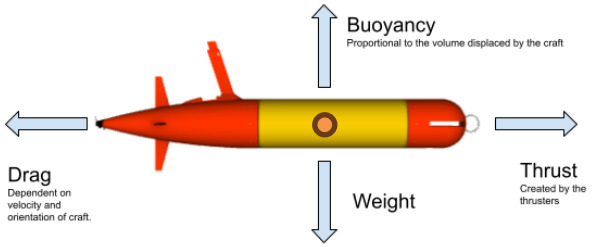
HOW?

- Continuous solver tools (such as Simulink)
- Simplifications
- Discrete-time stepping

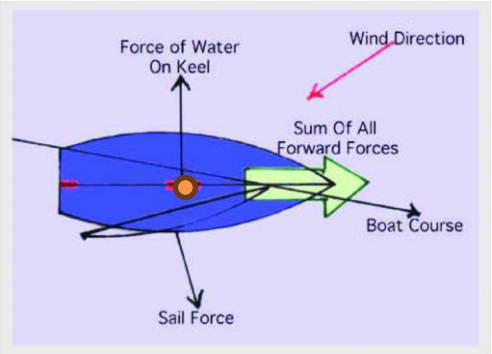
Forces and moments



Aerodynamics
Thermodynamics

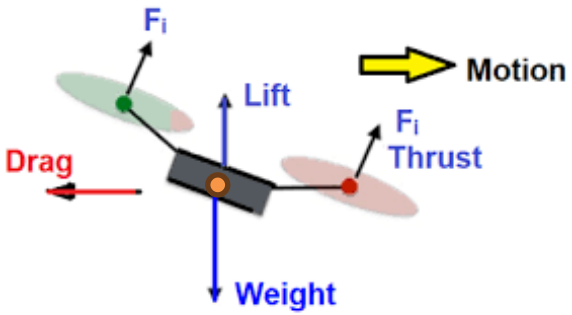


Hydrodynamics

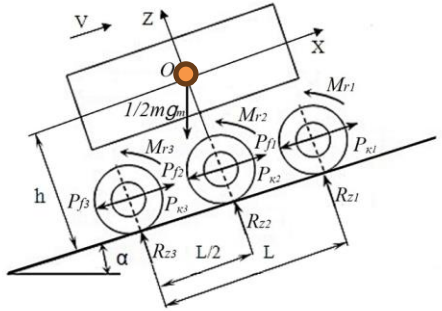


Aerodynamics
Hydrodynamics

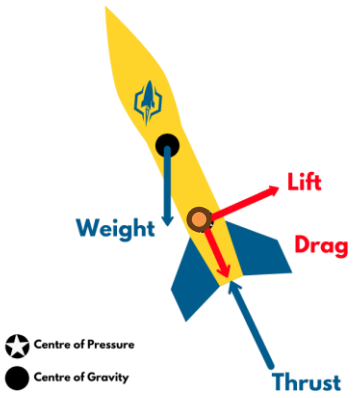
Forces and moments can be generated by many different physical phenomena



Propeller thrust
Aerodynamic drag

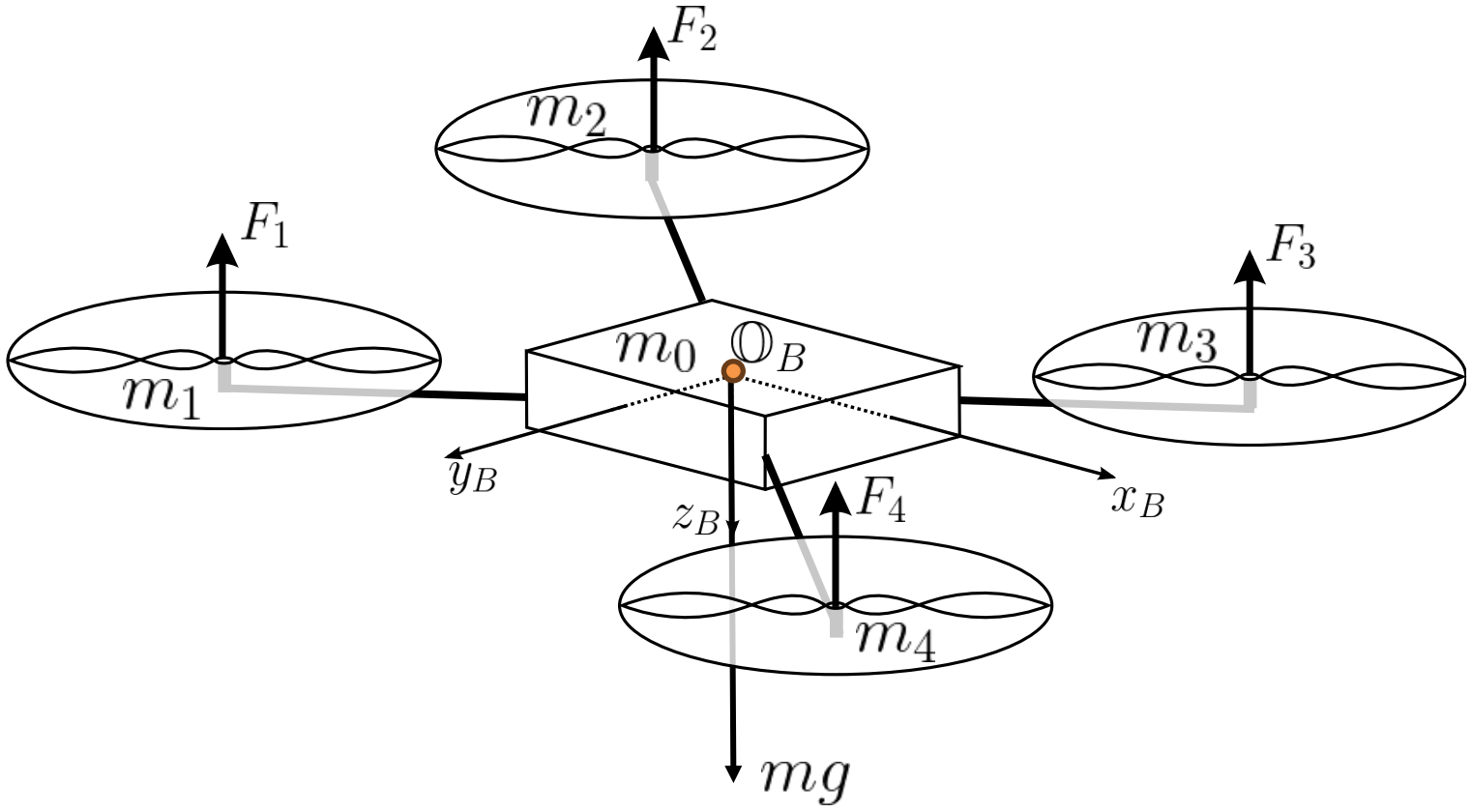


Friction
Mechanics



Thermodynamics

Example quadrotor…



Total mass is sum of sub-systems

$$m = \sum_{i=0}^{4} m_i$$

Total force is resolved sum of all contributions around c.o.m

$$X = \sum F_X$$

$$Y = \sum F_Y$$

$$Z = \sum F_Z$$

Rearranging for integral

The first step in using the rigid-body-dynamics equations is to rearrange for the derivatives

Translational

$$X = m(\dot{U} - rV + qW)$$
$$Y = m(\dot{V} - pW + rU)$$
$$Z = m(\dot{W} - qU + pV)$$

$$\dot{U} = \frac{X}{m} + rV - qW$$

$$\dot{V} = \frac{Y}{m} + pW - rU$$

$$\dot{W} = \frac{Z}{m} + qU - pV$$

Rearranging for integral

The first step in using the rigid-body-dynamics equations is to rearrange for the derivatives

Rotational – not so easy to invert for $\dot{p}, \dot{q}, \dot{r}$

$$\bar{M} = I\dot{\Omega} + (\Omega \times (I\Omega))$$

Assume cross inertias are zero

$$I = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$

Rearranging for integral

The first step in using the rigid-body-dynamics equations is to rearrange for the derivatives

Rotational (with assumption)

$$L = I_x\dot{p} - (I_y - I_z)qr \cancel{+ I_{xy}(pr - \dot{q}) - I_{xz}(pq + \dot{r}) + I_{yz}(r^2 - q^2)}$$
$$M = I_y\dot{q} + (I_x - I_z)pr \cancel{- I_{xy}(qr - \dot{p}) + I_{xz}(p^2 - r^2) + I_{yz}(pq - \dot{r})}$$
$$N = I_z\dot{r} - (I_y - I_y)pq \cancel{+ I_{xy}(q^2 - p^2) + I_{xz}(qr + - dotp) + I_{yz}(pr - \dot{q})}$$

$$\dot{p} = I_{xx}^{-1}\left[L + \left(I_{yy} - I_{zz}\right)qr\right]$$

$$\dot{q} = I_{yy}^{-1}[M + (I_{zz} - I_{xx})rp]$$

$$\dot{r} = I_{zz}^{-1}\left[N + \left(I_{xx} - I_{yy}\right)pq\right]$$

State representation

State representation

$$\dot{s} = f(s) + g(u)$$

Rate of change of state = function of current state and external inputs

$$s = \int_0^t \dot{s}$$

All in body frame! – must rotate via transformation to know absolute position and orientation in an external reference frame

$$u = [X, Y, Z, L, M, N]^T$$

External inputs are the summed forces and torques acting around centre of mass

Transformation of derivatives

Translational

$$\bar{b} = \mathbb{R}_{ab}\bar{a}$$

$$\mathbb{R}_{ab} = f(\Theta_{ab}) = \begin{bmatrix} c\theta c\psi & c\theta s\psi & -s\theta \\ (-c\phi s\psi + s\phi s\theta c\psi) & (c\phi c\psi + s\phi s\theta s\psi) & s\phi c\theta \\ (s\phi s\psi + c\phi s\theta c\psi) & (-s\phi c\psi + c\phi s\theta s\psi) & c\phi c\theta \end{bmatrix}$$

Rotation of linear vector $\bar{a}$ from Earth to Body frame

Apply inverse to convert linear velocity vector (U,V,W) from body frame to Earth frame

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \mathbb{R}_{ab}^{-1} \begin{bmatrix} U \\ V \\ W \end{bmatrix}$$

$$\mathbb{R} = f(\phi, \theta, \psi)$$

Transformation of derivatives

Rotational

There is a variation of the linear rotational matrix for rotating rotation rates.

See Cook, “Flight Dynamics Principals” (3rd Edition) Section 2.4.2 for derivation

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & sin\phi tan\theta & cos\phi tan\theta \\ 0 & cos\phi & -sin\phi \\ 0 & sin\phi/cos\theta & cos\phi/cos\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

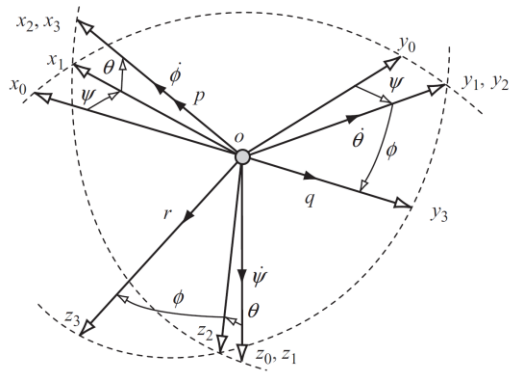


Figure 2.7 *Angular rates transformation.*

Diagram from Cook

2nd Integral

- The translational and rotational rates are now expressed in the Earth (external) reference frame
- Integrate to calculate the absolute position and orientation

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \int_0^T \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}$$

$$\begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix} = \int_0^T \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

Conceptually this is the 2nd integral to satisfy the relationship between force and position according to:

$$F = ma$$

$$T = I\dot{\omega}$$

Overview of all equations

1. $$[X \quad Y \quad Z \quad L \quad M \quad N]^T = f(x, y, z, \phi, \theta, \psi, U, V, W, p, q, r, u)$$

Forces and torques depend on many physical phenomena which are dependent on the state of the system and external command inputs

2. $$\dot{U} = Xm^{-1} + rV - qW \qquad \dot{p} = I_{xx}^{-1}[L + (I_{yy} - I_{zz})qr]$$
$$\dot{V} = Ym^{-1} + pW - rU \qquad \dot{q} = I_{yy}^{-1}[M + (I_{zz} - I_{xx})rp]$$
$$\dot{W} = Zm^{-1} + qU - pV \qquad \dot{r} = I_{zz}^{-1}[N + (I_{xx} - I_{yy})pq]$$

Body frame rates are calculated from rigid body dynamics and kinematic relationships.

3. $$\begin{bmatrix} U \\ V \\ W \end{bmatrix} = \int_0^t \begin{bmatrix} \dot{U} \\ \dot{V} \\ \dot{W} \end{bmatrix} \qquad \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \int_0^t \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix}$$

Integrate for body frame rates

4. $$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \mathbb{R}_{ab}^{-1} \begin{bmatrix} U \\ V \\ W \end{bmatrix} \qquad \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & sin\phi tan\theta & cos\phi tan\theta \\ 0 & cos\phi & -sin\phi \\ 0 & sin\phi/cos\theta & cos\phi/cos\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

Transform to external reference axes

5. $$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \int_0^T \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} \qquad \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix} = \int_0^T \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

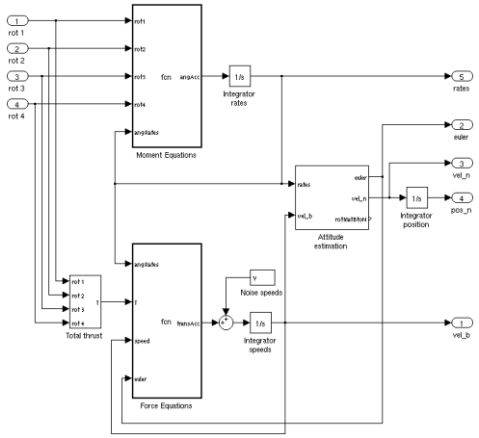
Integrate for external reference frame pose (position and orientation)

State = $(x, y, z, \phi, \theta, \psi, U, V, W, p, q, r, u)$

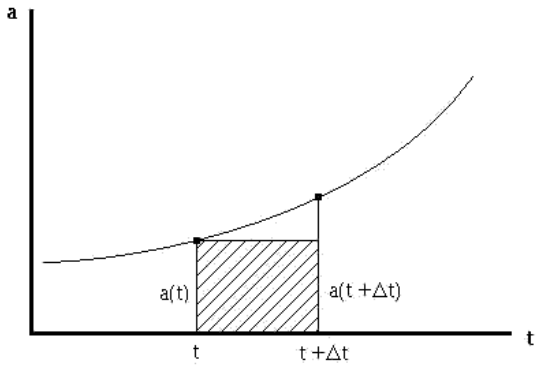
Resulting system state used to calculate new feedback control, forces, and torques

Solving the system of equations

- Many of the equations are circularly dependent on each other
  - i.e. the output of an equation is also an input at the same instant in time
- They form a set of non-linear ordinary differential equations
- Not possible to solve analytically or to iterate in the current form
- Solve by using a tool (Simulink), or direct discretisation via approximations



An example Simulink block diagram



A representation of the forward Euler integration method

Simulink

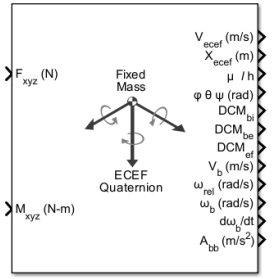
- See example on Blackboard
- Can download and run on any installation of MATLAB with Simulink
- open('AERO60492_AMR_RigidBodyDynamics’)
- Click ‘run’

to simulate the model
- Double click scope blocks

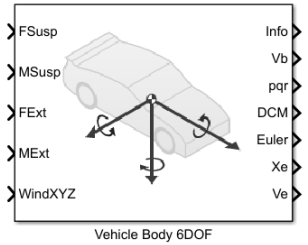
to view state time history

Library implementations

Aerospace blockset

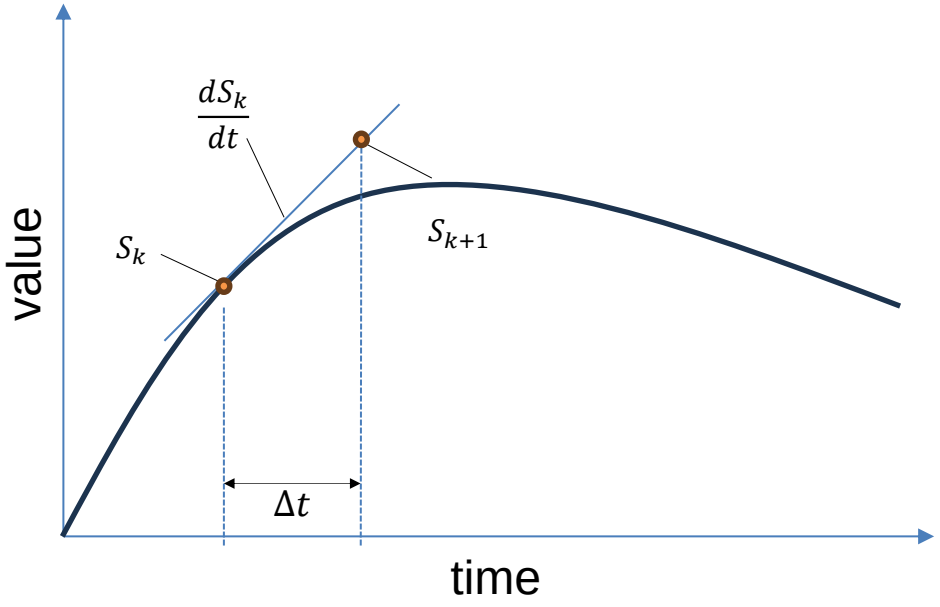


Vehicle dynamics blockset



Discrete time-stepping

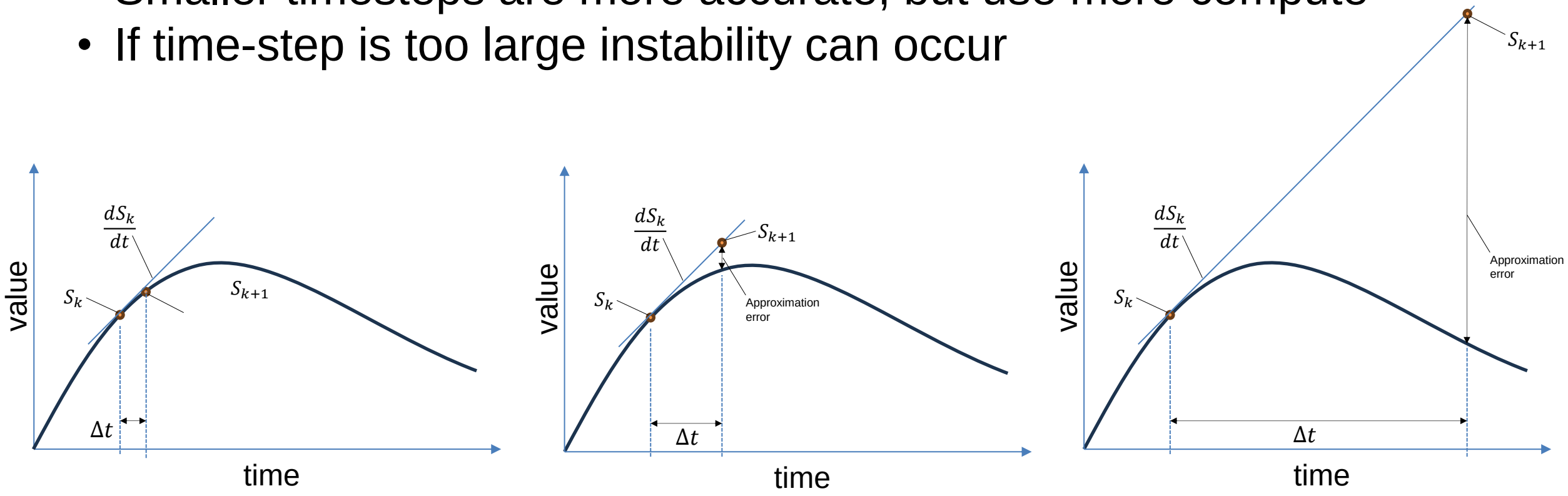
- The continuous-time differential equations and integral blocks cause the problem with direct iteration of equations
  - (see steps 2 and 5)
- Integrals can be solved using numerical methods
  - Simulink handles this using more advanced methods
- Most fundamental method is forward Euler



$$S_{k+1} = S_k + \Delta t \frac{dS_k}{dt}$$

Discrete time-stepping

- Euler method is an approximation
- Smaller timesteps are more accurate, but use more compute
- If time-step is too large instability can occur



Discrete time-stepping

Used to solve equations in step 2 & 3

Continuous time equations

$$\dot{U} = Xm^{-1} + rV - qW$$

$$\begin{bmatrix} U \\ V \\ W \end{bmatrix} = \int_0^t \begin{bmatrix} \dot{U} \\ \dot{V} \\ \dot{W} \end{bmatrix}$$

Forward Euler approximation

$$U_{k+1} = U_k + \Delta t \frac{dU_k}{dt}$$

$$\therefore\ U_{k+1} = U_k + \Delta t \dot{U}_k$$

Substitute full expansion for discrete-time equation

$$U_{k+1} = U_k + \Delta t[X_k m^{-1} + r_k V_k - q_k W_k]$$

An identical approximation can be applied to $[U, V, W, p, q, r]$ states and derivatives

Discrete time-stepping

Used to solve equations in step 4 & 5 and in vector notation

Continuous time equations

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \mathbb{R}_{ab}^{-1} \begin{bmatrix} U \\ V \\ W \end{bmatrix} \qquad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \int_0^T \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}$$

Forward Euler approximation

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{k+1} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_k + \Delta t \frac{d \begin{bmatrix} x \\ y \\ z \end{bmatrix}_k}{dt}$$

$$\therefore \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{k+1} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_k + \Delta t \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}_k$$

Substitute full expansion for discrete-time equation

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{k+1} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_k + \mathbb{R}_{ab}^{-1} \begin{bmatrix} U \\ V \\ W \end{bmatrix}_k$$

Iterative time-stepping rigid body dynamics equations

1. $$[X \quad Y \quad Z \quad L \quad M \quad N]_k^T = f[x, y, z, \phi, \theta, \psi, U, V, W, p, q, r, u]_k$$

2. $$\dot{U}_k = \{Xm^{-1} + rV - qW\}_k \qquad \dot{p} = \{I_{xx}^{-1}[L + (I_{yy} - I_{zz})qr]\}_k$$
$$\dot{V}_k = \{Ym^{-1} + pW - rU\}_k \qquad \dot{q} = \{I_{yy}^{-1}[M + (I_{zz} - I_{xx})rp]\}_k$$
$$\dot{W}_k = \{Zm^{-1} + qU - pV\}_k \qquad \dot{r} = \{I_{zz}^{-1}[N + (I_{xx} - I_{yy})pq]\}_k$$

3. $$\begin{bmatrix} U \\ V \\ W \end{bmatrix}_{k+1} = \begin{bmatrix} U \\ V \\ W \end{bmatrix}_k + \Delta t \begin{bmatrix} \dot{U} \\ \dot{V} \\ \dot{W} \end{bmatrix}_k \qquad \begin{bmatrix} p \\ q \\ r \end{bmatrix}_{k+1} = \begin{bmatrix} p \\ q \\ r \end{bmatrix}_k + \Delta t \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix}_k$$

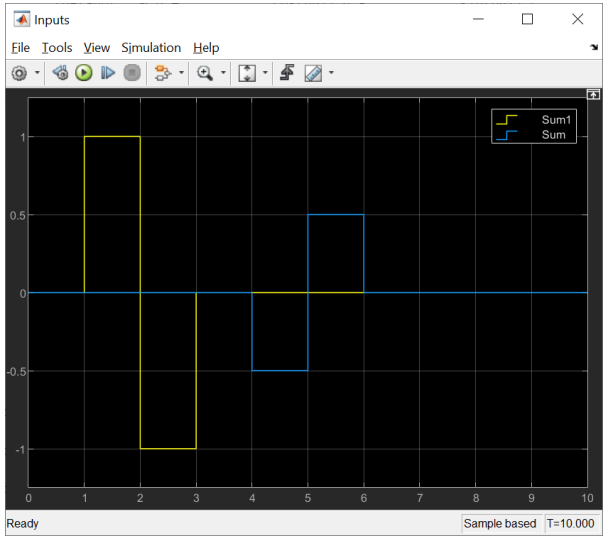
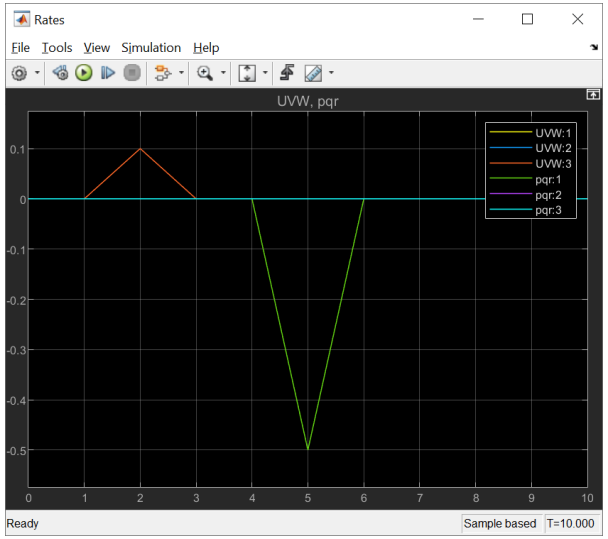
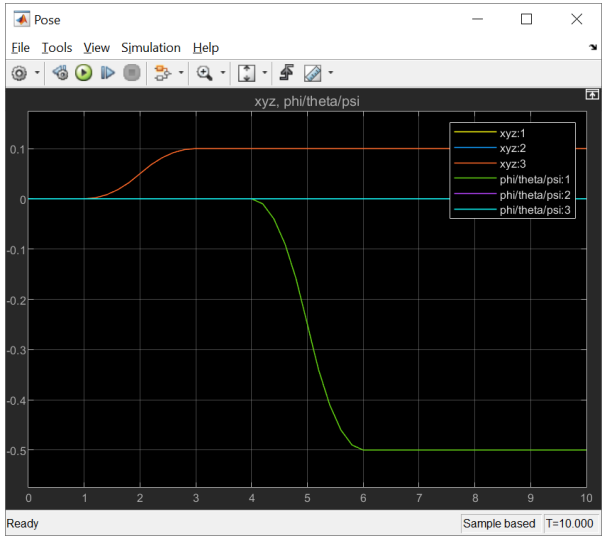
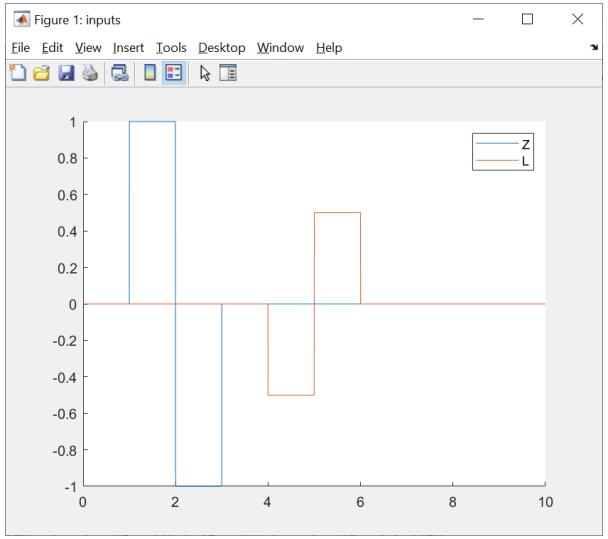
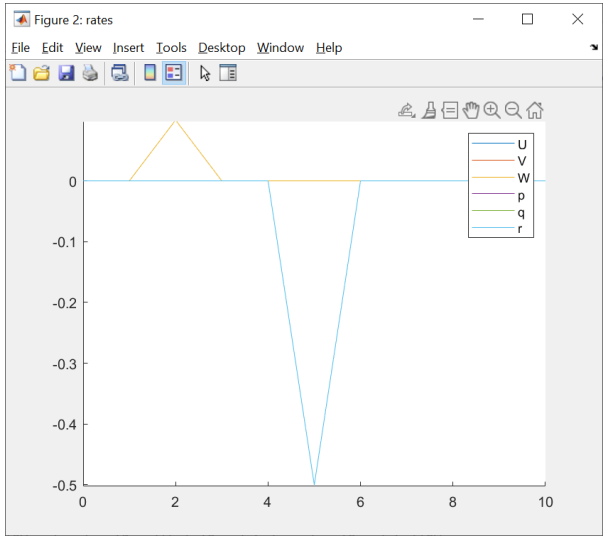
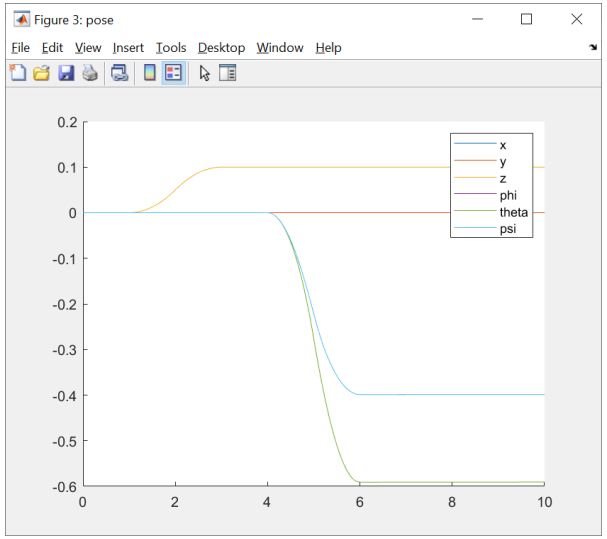
4. $$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \mathbb{R}_{ab}^{-1} \begin{bmatrix} U \\ V \\ W \end{bmatrix} \qquad \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & sin\phi tan\theta & cos\phi tan\theta \\ 0 & cos\phi & -sin\phi \\ 0 & sin\phi/cos\theta & cos\phi/cos\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

5. $$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{k+1} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_k + \Delta t \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}_k \qquad \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}_{k+1} = \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}_k + \Delta t \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}_k$$

State $=[x, y, z, \phi, \theta, \psi, U, V, W, p, q, r]_{k+1}$

$\{\quad\}_k$ shorthand used to indicate function should be evaluated using current time-step state values

Output comparison

Simulink

Discrete